

$$\therefore \theta = 20^\circ = 37.653^\circ \Rightarrow \theta = 77.653^\circ \approx 78^\circ$$

- 14 D** A stationary wave forms between G and H. As such the phase difference between E and F is π as they are in adjacent loops. As the frequency and period of points E and F are the same, F has to travel faster and hence possesses a higher maximum speed and thus maximum kinetic energy as compared to E.

- 15 A** 1